Engineering Challenge 2026

Politecnico di Milano — Autonomous Vehicle Competition Team

Scenario

Congratulations! You have just joined the autonomous vehicle competition team at Politecnico di Milano. To evaluate your technical skills, your team leader has assigned you a challenge based on real data recorded during an autonomous racing session on a track.

You will work with a ROS bag containing data recorded by an autonomous vehicle. The dataset includes:

- LiDAR data
- Radar data
- Camera streams
- Ego vehicle speed and odometry

In addition to technical skills, this challenge will also evaluate:

- Problem solving
- Work organisation
- Teamwork
- Quality of the final presentation
- Ability to deliver a clear and effective demo

Objective

Develop a perception and tracking system for autonomous racing using the data provided in the ROS bag. The system must process multi-sensor data to:

- Detect opponent vehicles
- Estimate their state and motion
- Maintain tracking over time
- Build a live map
- Predict future behaviour
- Dynamically represent the surrounding environment

Technical Requirements

- Any programming language may be used.
- Any framework or library may be used.
- Code must be runnable and documented.

Tasks & Scoring

Tasks are ordered by increasing difficulty. Each task contributes to the final score.

Core Tasks

Task	Points	Description	Evaluation criteria
Task 1 Opponent Detection	**15 pt**	Detect opponent vehicles using one or more sensors. • Target position • Bounding boxes / cluster detection • Visualisation of detected targets	• Accuracy • Robustness • False positive reduction
Task 2 Sensor Fusion	**25 pt**	Fuse data from Radar, LiDAR, and cameras to improve target state estimation. • Position • Velocity • Heading of opponent vehicle	• Estimation accuracy • Noise robustness • Handling of missing data
Task 3 Continuous Target Tracking	**30 pt**	Maintain consistent tracking of opponent vehicles over time. • Handle temporary occlusions • Maintain consistent identities • Avoid target loss	• Tracking continuity • Stability • Tracking accuracy

Bonus Tasks

Task	Points	Description	Evaluation criteria
Task 4 Environment Mapping	**+10 pt**	Build a dynamic representation of the surroundings. • Occupancy grid • Bird-eye view map • Free-space estimation • Track boundary reconstruction	• Map quality • Dynamic update • Visualisation clarity
Task 5 Trajectory Prediction	**+15 pt**	Predict the future trajectory of opponent vehicles. • Short-term prediction • Medium-term prediction	• Prediction accuracy • Stability • Handling of dynamic manoeuvres
Task 6 Behaviour Prediction	**+10 pt**	Estimate the intent of opponent vehicles.	• Classification correctness • Robustness

Task	Points	Description	Evaluation criteria
		• Lane keeping • Braking • Overtaking attempt • Trajectory recovery	
Task 7 Overtake Strategy / Planning	**+15 pt**	Develop a planning strategy for the ego vehicle. • Evasive trajectory • Overtaking strategy • Collision avoidance • Racing line adaptation	• Innovation • Safety • Strategy effectiveness

Final Deliverables

Each team must submit:

1. Source Code

Complete, runnable, and documented code. Any language, framework, or library is allowed.

2. Final Presentation

A PowerPoint presentation covering:

- Project overview
- Software architecture
- Chosen approach
- Algorithms used
- Developed pipeline
- Results obtained
- Problems encountered
- Team organisation
- Future improvements

3. Demo (nice to have)

A live or recorded demonstration of the solution, for example:

- Tracking visualisation
- Environment map
- Trajectory prediction
- Strategy simulation

Evaluation Criteria

Area	Weight
Technical quality	**40%**
Accuracy & robustness	**25%**
Software architecture	**10%**
Innovation	**10%**
Presentation & demo	**15%**

Good luck and enjoy the challenge! 🚙 🐦